1. Construct and communicate explanations and arguments in the context of the problem (Point 7)

Analysis: This is the **most central** requirement of the task. The prompt explicitly asks students to "Explain why this statement is incorrect" and "Give a reason." Success depends on the student's ability to construct a logically sound argument that points out the flaw in Kevin's reasoning and communicate it clearly (p. 1).

2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem (Point 2)

Analysis: The first step to solving this is recognizing that Kevin's conclusion is **unreasonable**. Students must compare the data—France has 24.5 million TV households while Norway has only 2.0 million—and realize that because the "base" population of France is over 12 times larger, Kevin's conclusion is mathematically impossible (p. 1).

3. Understand the relationship between the context of the problem and representation of the mathematical solution (Point 9)

Analysis: This item tests the understanding of the relationship between "relative values" (percentages) and "absolute values" (totals). Students must use this understanding to interpret the solution and point out the **limitations** of comparing percentages without considering the underlying base numbers.

4. Interpret mathematical outcomes in a variety of formats; compare or evaluate two or more representations (Point 5)

Analysis: Students must navigate and evaluate multiple data representations in the table: country names, absolute numbers (in millions), and two distinct percentage columns. The key is evaluating and integrating the "Number of households" column with the "Cable TV percentage" column, rather than focusing on a single representation in isolation (p. 1).

5. Use knowledge of how the real world impacts outcomes... to make contextual judgments (Point 6)

Analysis: Students use their real-world knowledge (e.g., knowing France is a much larger nation than Norway) to guide their mathematical judgment. This context helps them immediately sense the massive scale difference, allowing them to adjust their judgment and correctly identify why Kevin's logic is flawed (p. 1).

1. 在问题的背景下构建并传达解释和论点（第 7 项）

分析： 这是本题**最核心**的要求。题目明确要求“解释为什么这个陈述是错误的”并“给出理由”。学生必须构建一个逻辑严密的论点，指出凯文忽略了“基数”（电视家庭总数）的巨大差异，并清晰地表达出来。 (p. 1)

2. 识别数学结果或结论在给定问题背景下是否合理（第 2 项）

分析： 解决此题的第一步是意识到凯文的结论是**不合理**的。学生需要通过比较数据（法国 2450 万 vs 挪威 200 万）发现，尽管挪威的订购率（42.7%）高于法国（15.4%），但由于法国的总量是挪威的 12 倍以上，凯文的结论在逻辑上站不住脚。 (p. 1)

3. 理解背景与数学表示之间的关系，并评估其局限性（第 9 项）

分析： 此题考察了对“相对量（百分比）”与“绝对量（具体数值）”这两种数学表示之间关系的深度理解。学生需要指出，仅凭百分比这一表示方式在比较不同规模的实体（国家）时是有局限性的，必须结合基数才能得出正确结论。 (p. 1)

4. 解释多种格式的数学成果，并比较或评估两种或多种表示（第 5 项）

分析： 学生必须在表格中处理和比较多个维度的信息：不同国家的名称、家庭绝对数量（百万）以及两个不同的百分比列。解题关键在于评估并整合“电视家庭数量”列与“有线电视百分比”列，而不是像凯文那样只关注其中一列。 (p. 1)

5. 利用现实世界对结果的影响做出背景判断（第 6 项）

分析： 学生需要利用常识或背景知识（法国是一个比挪威大得多的国家）来辅助判断。这种现实世界的规模感有助于学生迅速察觉计算中的数量级差异，从而对凯文的判断做出修正。 (p. 1)